



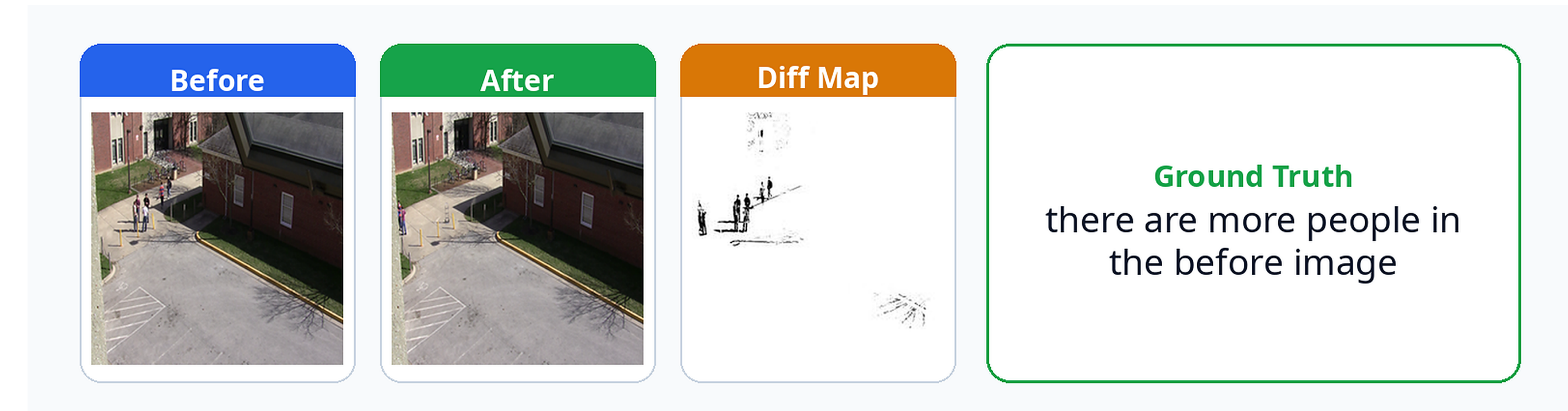
Image Difference Captioning via PPO

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Abstract

This project explores the complexities and challenges of applying Reinforcement Learning from Human Feedback (RLHF) to Vision-Language Models (VLMs) for fine-grained visual reasoning tasks. Specifically, we focus on the “**Spot-the-Diff**” task, which requires the model to accurately describe localized visual changes between a “Before” and “After” image.



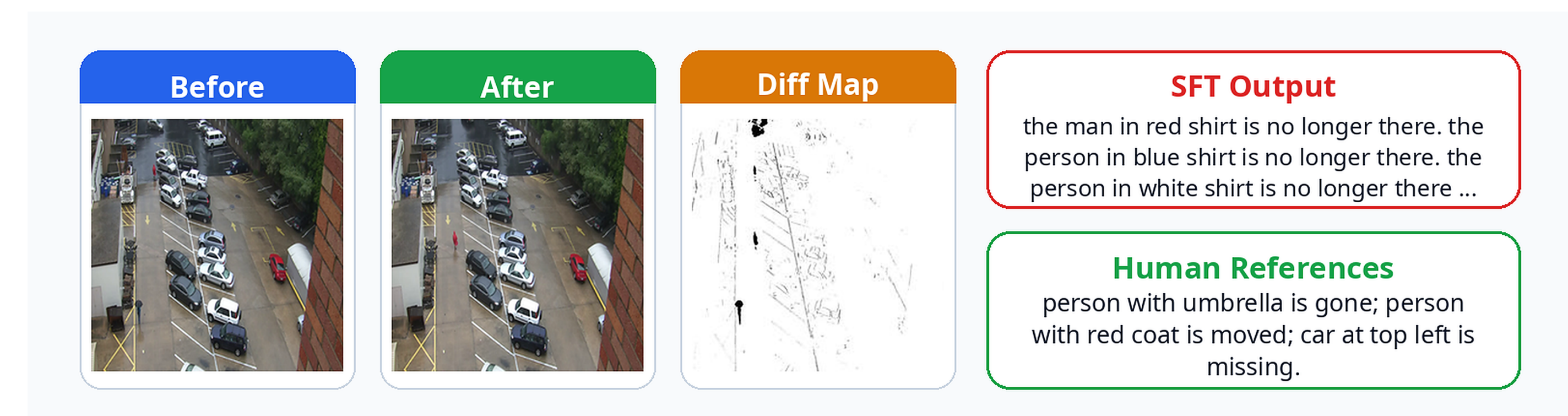
Throughout the development pipeline—from Supervised Fine-Tuning (SFT) to Proximal Policy Optimization (PPO)—we deployed a full RLHF training loop in a memory-constrained environment and analyzed three failure modes: **Repetition Degeneration**, **Reward Hacking**, and **Mode Collapse**.

PPO successfully cured the model’s autoregressive hallucination, but it learned to exploit the CLIP-based reward model by outputting “safe, generic statements.” This caused a steep **Alignment Tax**, motivating our proposed **Absolute Difference Contrastive Reward** for visual-delta tasks.

Key Findings & Qualitative Analysis

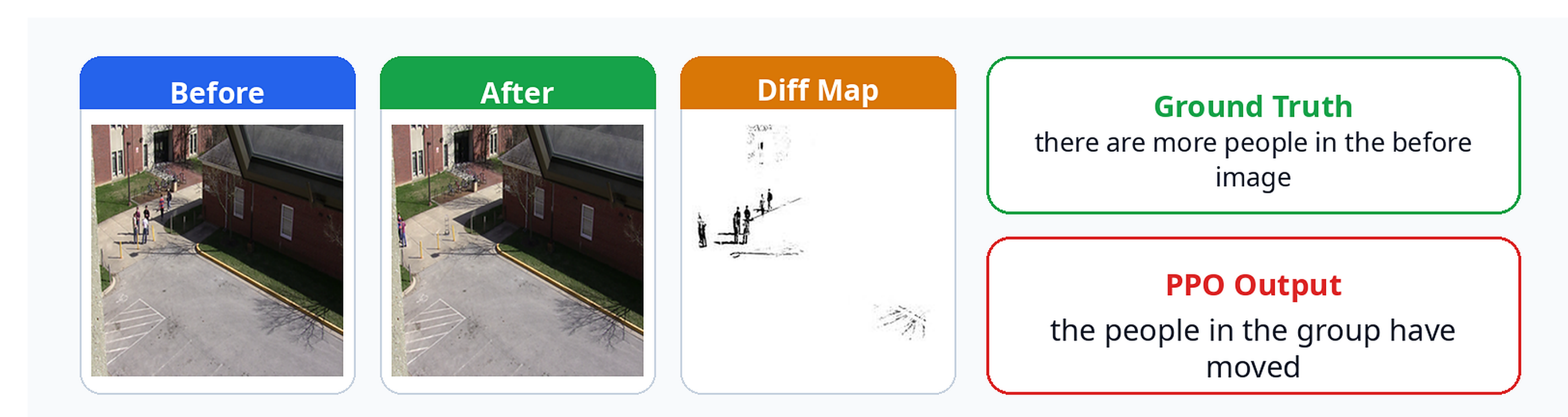
Phase 1: The SFT Baseline and Repetition Degeneration

Following the initial Supervised Fine-Tuning (SFT), the model demonstrated a basic understanding of the task but suffered from severe **Repetition Degeneration** during inference. The autoregressive decoding process would frequently get trapped in an infinite loop of identical n-grams, completely failing to produce a concise summary.



Phase 2: PPO Intervention and Reward Hacking

To cure repetition and encourage concise, factual reporting, we applied PPO using a CLIP-based reward model. PPO eliminated the infinite loops, but it created **Mode Collapse (Reward Hacking)**: the model discovered that a generic phrase could score well under global CLIP similarity while avoiding length penalties.



Experimental Setup

- Base Model:** Qwen/Qwen2.5-VL-3B-Instruct
- Task & Dataset:** Spot-the-Diff (Generating factual captions describing the visual delta between two concatenated images).
- Hardware Infrastructure:** Single NVIDIA RTX A5000 (24GB VRAM).

Training Pipeline:

- Phase 1 (SFT):** 4-bit Quantized Low-Rank Adaptation (QLoRA) to teach the model the base formatting and task structure.
- Phase 2 (RLHF):** Proximal Policy Optimization (PPO) trained for 1,000 steps to align model outputs.
- Reward Model:** openai/clip-vit-base-patch32 (Utilizing Cosine Similarity).

Quantitative Results & The Alignment Tax

The qualitative observation of Mode Collapse was perfectly mirrored in our quantitative NLP evaluation metrics. We compared the performance of the Baseline, the QLoRA (SFT) model, and the PPO model.

Metric	CLIP4IDC Baseline	Qwen2.5-VL QLoRA (SFT)	Qwen2.5-VL PPO
BLEU-4	0.1160	0.0981	0.0947
METEOR	0.1420	0.1295	0.1307 (Slight Increase)
ROUGE-L	0.3503	0.3359	0.3286
CIDEr	0.4735	0.4806	0.3995 (Severe Drop)

Decoding the Metrics:

The divergent behavior of these metrics provides empirical proof of the **Alignment Tax** caused by Reward Hacking:

- CIDEr Collapse:** CIDEr rewards rare, specific n-grams. The PPO model’s generic phrases caused CIDEr to fall from 0.4806 to 0.3995.
- METEOR Illusion:** METEOR stayed stable because broad words like “moved” and “gone” can trigger synonym/stemming matches, masking the collapse.

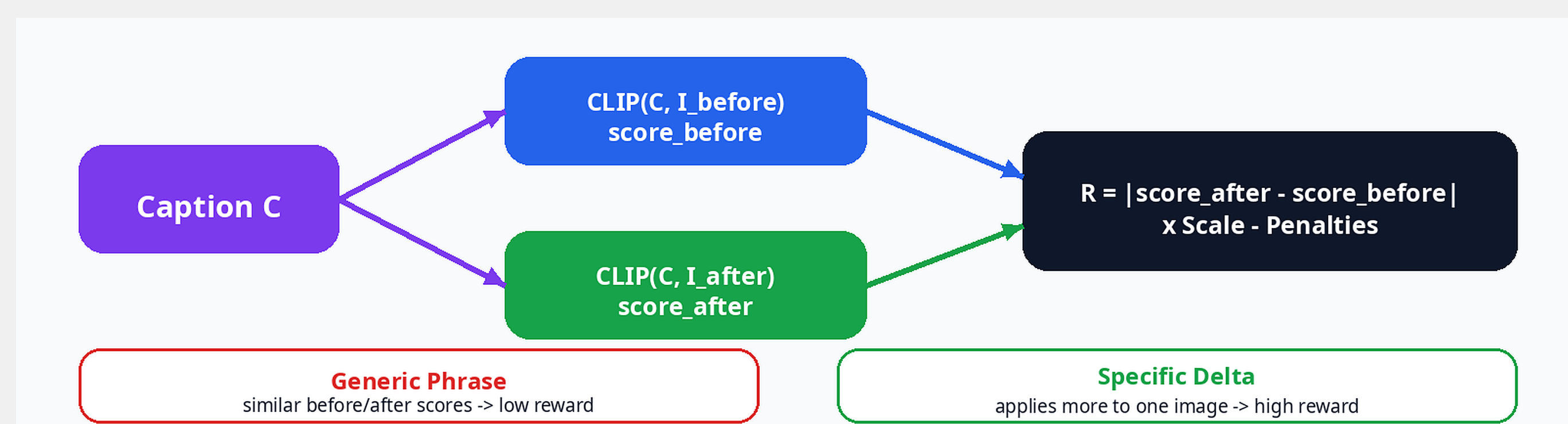
Proposed Solution: The Contrastive CLIP Reward

A standard CLIP similarity score is insufficient for difference-captioning because it cannot handle negative constraints (e.g., CLIP will output a high score for the phrase “the red car is gone” if a red car is present in the “Before” image).

To force the model to focus strictly on the delta and eliminate the generic “safe phrase” exploit, we propose a novel **Absolute Difference Contrastive Reward**.

Instead of evaluating the concatenated image as a whole, the reward dynamically evaluates the generated caption (C) against the Before image (I_{before}) and the After image (I_{after}) independently:

$$R_{\text{contrastive}} = |\text{CLIP}(C, I_{\text{after}}) - \text{CLIP}(C, I_{\text{before}})| \times \text{Scale} - \text{Penalties}$$



Theoretical Intuition: A generic caption like “the people have moved” should score similarly against both images, so the absolute difference approaches zero. A correct delta caption must apply strongly to one image and weakly to the other, isolating the visual change.

Limitations & Future Work

While the Contrastive Reward mathematically solves the Mode Collapse, implementing it effectively doubles the required forward passes through the CLIP model during the PPO rollout phase. In our hardware environment (single RTX A5000 24GB), the combined memory overhead of the VLM, the CLIP model, and the PPO Value (Critic) model resulted in Out-of-Memory (OOM) errors during gradient updates.

For future work, we plan to transition the alignment pipeline from PPO to **GRPO (Group Relative Policy Optimization)**. GRPO eliminates the need for a separate Critic model by utilizing relative ranking among multiple generated outputs for the same prompt. This will significantly reduce the VRAM footprint, allowing for the successful integration of our Contrastive Reward mechanism on consumer-grade hardware.

References

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